

Chunking Strategies

1.) Fixed Length Chunking

Chunks the data with predefined length.

Pros:

- * Easy to implement
- * Consistent chunk size; }
easy to index

Cons:

- * Context Loss
- * Relevance Issues.

Best for: FAQ, Simple docs

2.) Sentence Based Chunking:

Chunks the data on Sentence Based.

Pros:

- * Context preservation
- * Ease of Implementation

Cons:

- * Limited Context
- * Variable Length

Best for: Short or direct responses like customer queries / conversational AI.

3.) Paragraph Based Chunking:

Chunks the data based on paragraph

Pros:

- * Richer Context
- * Logical Division

Cons:

- * Inconsistent Sizes
- * Token Limits

Best for: Structured docs like articles, reports or documents

4.) Sliding Window Chunking:

Create a overlapping chunks by sliding a window thru the text.

Pros:

- * Context Continuity
- * Improved Retrieval

Cons:

- Redundancy
- Computational Cost

Best for: Docs where maintaining context across sections is critical.

5.) Semantic chunking:

Utilizes embedding or ML models to split the text based on semantic meaning, ensuring each chunk is cohesive in topic or idea.

Best for: Complex queries requiring deep understanding such as manual or academic papers.

Pros:

- * Contextual Relevance
- * Flexibility

Cons:

- * Complexity
- * Processing Time

6.) Recursive Chunking

Breaks down text progressively into smaller chunks using hierarchical delimiters like headings, subheadings, paragraphs & sentences

Pros:

- * Hierarchical Context
- * Scalability

Best for:

Large, hierarchical docs like books or extensive reports.

Cons:

- * Complex Implementation
- * Potential Context Loss if not managed small chunks properly

7.) Context Enriched chunks.

Enriches each chunks by adding summaries or metadata from surrounding chunks, maintaining context across sequences.

Best for: Long documents where the coherence across chunks is essential.

Pros:

- * Enhanced Context
- * Improved Coherence

Cons:

- * Complexity
- * Storage overhead

(8) Modality based chunking:

Handles different content types separately.

chunking each according to nature.

Pros:

- * Tailored Approach
- * Improved Accuracy

Cons:

- * Implementation Complexity
- * Integration Difficulty

Best for: Mixed Media docs like scientific paper or user manuals.

(9.) Agentic Chunking:

Employs a Large Language Model to analyse the text & suggest boundaries based on the content structure & semantics.

Best for: Complex documents where meaning & context is critical.

Pros: * Intelligent Segmentation
* Adaptive

Cons: * Computationally Intensive
* Cost

10.) Subdocument Chunking:

Summarizes the entire document or large sections and attach These summaries to individual chunks as a metadata.

Best for: Enhancing retrieval efficiency in extensive document collections.

Pros:
* Hierarchical retrieval
* Context Depth

Cons:
* Additional Processing
* Metadata Management

11.) Hybrid Chunking

Combines multiple chunking strategies to adapt dynamically to different query types or document structures.

Best for: Versatile systems handle different query types

Pros: * Flexibility
* Optimized Performance

Cons:
* Complex Logic for decisions
* Maintenance